

# SANJEEV KUMAR SETHI

NIT Surathkal '26 · Cloud-Native Networking Engineer (Go, eBPF/Cilium, Kubernetes)

[sanjeevsethilive@gmail.com](mailto:sanjeevsethilive@gmail.com) | +91-7975965223 | [sanjeevsethi.in](https://sanjeevsethi.in) | [linkedin.com/in/Sanjeevliiv](https://linkedin.com/in/Sanjeevliiv) | [github.com/Sanjeevliiv](https://github.com/Sanjeevliiv)

## EDUCATION

**National Institute of Technology Karnataka**  
*B.Tech. in Electrical & Electronics Engineering*

Mangaluru, Karnataka  
*Dec. 2021 – May 2026*

## EXPERIENCE

**Network Engineering Intern — NITK**

Jun. 2025 – Dec. 2025

- Built a **multi-WAN link aggregation** gateway on **OpenWrt** using **Multipath TCP (MPTCP)**, bonding **2** independent uplinks into a single logical connection.
- Benchmarked aggregate throughput at **70–80%** of the combined capacity of both links, roughly **1.5x** the throughput of a single uplink.
- Validated **fault tolerance** by failing one uplink mid-transfer; active **TCP** sessions failed over to the remaining link with **zero connection resets**.
- Deployed and tested the gateway on **Raspberry Pi** hardware in a lab environment under simulated real-world **Linux** network traffic.

## OPEN SOURCE CONTRIBUTIONS

**Cilium** (*eBPF-based Networking, Go*)

Feb. 2026

- Authored and Merged PR #44202: Migrated IP handling in the **nodemap** package from **net.IP** to **netip.Addr**, Eliminating heap allocations and ensuring immutable, Type-safe node IP representations.

## PROJECTS

**Layer 2 Packet Sniffer** (*Go, Linux, Raw Sockets*)

Aug. 2026 – Present

- Developed a link-layer packet sniffer in **Go** leveraging **AF\_PACKET** raw sockets to capture and inspect live Ethernet frames directly from network interfaces.
- Decoded **Ethernet headers**, extracting source/destination **MAC addresses**, **EtherType**, and encapsulating protocols (**IPv4**, **ARP**) for real-time traffic analysis.

**Site Reliability Engineering Platform** (*Go, Terraform, Kubernetes*)

Jan. 2026 – Apr. 2026

- Designed and deployed a microservices platform on **GKE Autopilot** using **Terraform**, ensuring reproducible infrastructure and eliminating configuration drift.
- Secured cluster networking with **Network Policies** (controlling **TCP/IP**, **DNS** traffic), private subnets (VPC), and non-root container enforcement.
- Designed **CI/CD** pipelines with **GitHub Actions**, achieving zero-downtime deployments via multi-stage Docker builds and Helm charts.
- Defined **SLIs**, **SLOs**, and **SLAs** targeting 99.9% availability.

## TECHNICAL SKILLS

**Languages:** Go, Python, Bash

**Networking:** TCP/IP, Raw Sockets (AF\_PACKET), MPTCP, DNS, HTTP, Routing, IPv4/IPv6

**Operating Systems:** Linux, Systemd, Process Management, Shell Scripting

**Cloud & DevOps:** Kubernetes, Docker, Terraform, Helm, CI/CD, GitHub Actions, GCP

**Observability & Reliability:** Prometheus, Grafana, SLIs, SLOs, SLAs

## CO-CURRICULAR ACTIVITIES

**Endurance Sports:** Finished 3 Half Marathons (21km each), demonstrating discipline and long-term perseverance.